MIL–P–8045B
10 APRIL 1964
SUPERSEDING
MIL–P–8045A
18 FEBRUARY 1955

MILITARY SPECIFICATION

PLASTIC, SELF-SEALING AND NON-SELF-SEALING TANK BACKING MATERIAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

* **1.1 Scope.** This specification covers requirements for plastic backing material (see 6.1).

* **1.2 Classification**

1.2 **Classes.** The plastic backing material shall be furnished in the following classes:

Class 1—For use with self-sealing and non-self-sealing fuel tanks.
Class 2—For use with self-sealing and non-self-sealing oil tanks.

1.2.2 *Types.* The plastic backing material shall be furnished in the following types:

Type I —Intermittently supported
Type II —Continuously supported
Type III—Intermittently supported and fastened to the supporting stringers

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P–D–680 —Dry Cleaning Solvent
QQ–A–355 —Aluminum Alloy Plate and Sheet, 2024
QQ–A–362 —Aluminum Alloy Plate and Sheet, Alclad 2024

MILITARY

MIL–S–3136—Standard Test Fluids, Hydrocarbons
MIL–T–5578—Tank, Fuel, Aircraft, Self-Sealing
MIL–O–6081—Oil, Lubricating, Jet Engine
MIL–T–6396—Tanks, Fuel, Oil, Water-Alcohol, Coolant Fluid, Aircraft, Non-Self-Sealing, Removable, Internal
MIL–P–7936—Parts and Equipment, Aeronautical, Preparation for Delivery

STANDARDS

FEDERAL

FED–STD–406—Plastics: Methods of Testing

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–129 —Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8030

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* **2.2 Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

* **3.1 Preproduction.** This specification makes provisions for preproduction inspection (see 4.3).

* **3.2 Material.** Unless otherwise approved by the procuring activity, the material shall consist of a specific plastic resin with suitable fillers and reinforcing agents, and shall conform to the applicable requirements specified herein.

3.3 Dimensions. Dimensions of sheets or panels including contoured parts shall be as specified in the contract, order, or applicable drawing. Tolerance shall be ± 1 percent.

* **3.3.1** *Thickness.* Unless otherwise specified, the variation in thickness shall not be greater than ± 10 percent from the specified thickness.

* **3.4 Weight.** The weight of the plastic backing material per square foot shall be as specified in the contract, order, or applicable drawing. Unless otherwise specified, the variation in weight per square foot shall not be greater than ± 10 percent.

3.5 Flammability. The burning rate shall not be greater than 2 inches per minute.

3.6 Water absorption. The maximum water absorption shall be 5 percent by weight, and the change in any linear dimension except thickness shall not be greater than 0.2 percent after 7 days water immersion.

* **3.7 Fuel and oil absorption.** The fuel absorption of the Class 1 material shall not be greater than 6 percent by weight after 7 days immersion in fuel conforming to Specification MIL–S–3136, Type III. The percent change in weight after 7 days immersion in oil conforming to Specification MIL–O–6081 shall not be greater than 6 percent by weight for Class 2 material.

3.8 Corrosion test. The material shall not be corrosive to aluminum alloys when tested in accordance with 4.6.4.

* **3.9 Tensile impact strength.** The minimum average tensile impact strength in both the lengthwise or crosswise direction shall be 12 foot-pounds per inch width of the gauge section. The average tensile impact strength when tested in the strongest direction at −65° ±4°F shall be at least equal to the value obtained in the strongest direction at standard conditions.

* **3.10 Tensile strength.** The minimum average as-received ultimate tensile strength in the lengthwise direction shall be 22,000 psi, and in the crosswise direction shall be 18,000 psi. The minimum average ultimate tensile strength for Class 1 plastic backing material after 8 days ±5 hours exposure to 160° ± 4°F, shall be 15,000 psi; and for Class 2 plastic backing material after 8 days ±5 hours exposure to 275° ±4°F, shall be 11,000 psi for either direction. The average ultimate tensile strength and secant modulus at 1 percent strain after 7 days ±5 hours immersion in water shall be at least two-thirds of the average as-received value for both lengthwise and crosswise directions.

* **3.11 Flexural strength for Type I material.** Type I material, Classes 1 and 2, shall be capable of supporting a concentrated load of 25 pounds per inch width at the center when loaded on a 2-inch span in both lengthwise and crosswise direction. Type I plastic backing material, Class 1, shall be capable of supporting a load of 10 pounds per inch width

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when tested at 160° ±4°F, after 8 days ±5 hours exposure to 160° ±4°F. Type I plastic backing material, Class 2, shall be capable of supporting a load of 10 pounds per inch width when tested at 275° ±4°F, after 8 days ±5 hours exposure to 275° ±4°F. Flexural strength shall be conducted according to 4.6.7.

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* **3.12 Identification of product.** Each production panel or sheet of backing material shall be permanently identified with a repetitive or intermittent strip form of marking to show the specification number, class, type, manufacturer's identification number, and order number. An arrow shall be used pointing in the longitudinal (machine or warp) direction. This information shall be legibly marked in characters not less than ⅜ inch in height using a suitable marking fluid so that the markings produced shall not be removable by water, gasoline, or lubricating oil. The characters shall be applied in rows of constantly recurring symbols from one edge of the sheet or panel to the opposite edge. The marking shall be such that no piece larger than 18 inches square can be cut from the sheet or panel without bearing the above information.

* **3.13 Workmanship.** The plastic backing material shall be uniformly constructed, free from holes, cracks, voids, blisters, bubbles, wrinkles, folds, pockets of free resin or solvent, dirt, foreign material, rough or uneven surfaces or edges, delaminations, and shall be uniform in texture and color.

4. QUALITY ASSURANCE PROVISIONS

* **4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* **4.2 Classification of inspection.** The inspection of the plastic backing material shall be classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

* **4.3 Preproduction inspection.** Preproduction inspection shall consist of all the inspection procedures of this specification (see 4.6).

4.3.1 *Sampling instructions.* The preproduction samples shall consist of three panels of backing material, each at least 24 inches by 30 inches.

4.3.2 *Preproduction tests.* The mechanical and physical tests listed in Table II shall be conducted on the three sample panels selected in 4.3.1. Specimens shall be cut to size for all tests as specified in Table I.

* **4.4 Quality conformance inspection.** Quality conformance inspection shall consist of all examination procedures of this specification (see 4.4.3) and the test for ultimate tensile strength in the as-received condition (see 4.4.4, 4.6.6).

4.4.1 *Sampling for inspection and acceptance.* Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Standard MIL–STD–105. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same class and type submitted for delivery at one time.

4.4.2 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

** Gunfire requirements have been deleted.

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4.4.3 *Inspection of the end item*

4.4.3.1 *Examination of the end item.* Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL–STD–105, shall be expressed in units of sheets or contoured shapes for examinations in 4.4.3.1.1, 4.4.3.1.2, and in units of shipping containers for examination in 4.4.3.1.3.

TABLE I. *Sample size and number*

Item	Tests	Minimum number of specimens required	Size of specimen
1	Thickness	3	24 in. by 30 in.
2	Weight	3	Use Item 1 panels
3	Flammability	5	6 in. by ½ in.
4	Water absorption	5	1 in. by 3 in
5	Fuel absorption	5	1 in. by 3 in.
6	Corrosion	6	2½ in by 2½ in.
7	Tensile impact	18	See 4.6.5
8	Tensile	30	Type I in Standard FED–STD–406 (except as noted in 4.6.6)
9	Flexural	24	1 in. by 4. in

TABLE II. *Mechanical and physical test methods*

Tests	Standard FED-STD-406 method No. or specification paragraph No.
Flammability	2021
Water absorption	7031 Procedure A (except immersion shall be 7 days)
Fuel absorption	7031 Procedure A (except immersion shall be 7 days in test fuel or oil)
Corrosion	4.6.4
Tensile impact	4.6.5
Tensile	4.6.6
Flexural	4.6.7

4.4.3.1.1 *Examination of the end item for defects in appearance, workmanship, and markings.*

Examine	*Defect*
Appearance	Holes, cracks, voids, blisters or bubbles.
	Wrinkles or folds.
	Pockets of free resin or solvent.
	Presence of dirt, foreign material.
Workmanship	Rough or uneven surfaces or edges.
	Delaminations.
	Not uniform texture and color.
	Material not as specified.
Markings	Item not correctly or completely identified as required (see 3.12).
	Not legible, less than 5/32 inch in height.
	Spacing more than 18 inches apart.
	Marking ink affected by water, gasoline or oil.*

* Contractor shall furnish a certificate of compliance indicating conformance.

4.4.3.1.2 *Examination of end item for dimensional and weight defects.* The sample unit for this examination shall be one sheet or contoured shape.

Examine	*Defect*
Length and Width	Unless otherwise specified varies by more than ±1 percent from specified or required dimension.
Thickness	Thickness varies by more than ±10 percent from the specified thickness, unless otherwise specified.
Contoured shapes	Not as specified in contract, order, or applicable drawing.
Weight per square foot	Varies by more than ±10 percent from the specified nominal value (weight to the nearest 0.01 pound).

4.4.3.1.3 *Examination of preparation for delivery.* An examination shall be made to determine that packing, contents, and markings comply with the requirements of Section 5 of this specification. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the

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closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	*Defect*
Packing	Container material not as specified.
	Not level specified; not in accordance with contract requirements.
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability.
	Closure not accomplished by specified or required methods or materials.
	Sheets or molded parts not interleaved.
Count	Less than indicated quantity of sheets or contoured shapes per shipping container.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

4.4.3.1.4 *Inspection levels and acceptable quality levels (AQL's) for examination.* The inspection levels for determining the sample size, and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Examination Paragraph	*Inspection level*	*AQL*
4.4.3.1.1	I	2.5
4.4.3.1.2	S–3	4.0
4.4.3.1.3	S–2	4.0

4.4.4 *Lot acceptance testing.* Lot acceptance testing of the end item shall be conducted in accordance with 4.6.6 for each lot presented for inspection. The sample unit shall be a panel at least 2-½ feet by 2-½ feet. The sample size shall be S–1 except that not less than three (3) sample units shall be randomly selected throughout the lot and the acceptance number shall be zero. The lot size shall be expressed in units of plastic sheets or contoured shapes of the same type or class. There shall be no failure of the average values of any sample unit to meet the applicable requirement (ultimate tensile strength of "as received" material) specified in 3.10.

* 4.5 **Inspection conditions.** All specimens in Table I shall be conditioned and tested at standard conditions unless otherwise specified herein or in Standard FED–STD–406. Standard conditions shall be 23° ± 1.1°C (73.5° ± 2°F) and 50 ± 4 percent relative humidity. Specimens shall be tested after being exposed for 4 days to this temperature and humidity.

* **4.6 Inspection procedures**

4.6.1 *Thickness test.* Thickness shall be measured with micrometer calipers around the edges of the panels. At least eight measurements to the nearest 0.001 inch shall be made on each panel.

4.6.2 *Weight.* Weight per square foot shall be determined to the nearest 0.01 pound for each panel.

4.6.3 *Mechanical and physical properties.* The mechanical and physical properties of the laminates shall be determined in accordance with Table II.

4.6.4 *Corrosion test.* The specimens shall be prepared by cutting six strips of backing material 2-½ inches by ½ inch and cleaned by scrubbing with soap and water, rinsed in distilled water and 95 percent methanol, and air dried. The material shall be suspended in a controlled atmosphere of 95 percent relative humidity at 100° ± 2°F, approximately 16 hours prior to the corrosion test. Test panels measuring 2 inches by 2 inches by ⅛ inch shall be made from aluminum alloy conforming to Specification QQ–A–362 and aluminum alloy conforming to Specification QQ–A–355, degreased by rinsing in hot naphtha conforming to Specification P–D–680, and finished by rubbing with number 240 or 280 alundum abrasive cloth or paper until a uniform surface is obtained. Prior to use, the panels shall be cleaned by swabbing and rinsing in naphtha, air dried, and finally rinsed by immersing in boiling 95 percent and 100 percent naphtha, respectively. In the test procedure, two of the prepared specimens shall be placed between two of the prepared clad aluminum test panels. These two "sandwiches" shall then be placed together with two more test specimens placed between

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them. A compressive force of 40 to 50 pounds per square inch of the test specimens or a total of 100 to 125 pounds shall be applied to the aluminum plates by a spring-loaded clamping jig. The test assembly carrying the test specimens shall be placed in a controlled atmosphere of 95 percent relative humidity at 100° ± 2°F for 7 days. Corrosion other than staining and very light pitting shall be cause for rejection of the material if confirmed by retest. Final evaluation will be made by the procuring activity.

* 4.6.5 *Tensile impact tests.* Tensile impact specimens shall be at least 9 inches by 1 inch in overall size and shall have a configuration similar to the flat tension specimen of Type I as specified in Method 1011 of Standard FED–STD–406. To insure failure of the specimen in the gage section, this section may be 0.40 inch wide instead of the specified 0.50 inch. Each grip end of the specimen shall have two bolt holes, each no larger than ¼ inch in diameter, spaced at least 1 inch apart and ½ inch from the end along the longitudinal axis of the specimen. Tests shall be conducted on a pendulum-type impact machine. The specimen shall be held in rough surfaced grips with two bolts through each end (see Figure 1). The specimen shall be mounted so that one grip is stationary and the other is free to move in a direction along the longitudinal axis of the specimen. The free grip-end shall strike the anvil of the testing machine with sufficient force to break the specimen. Tests shall be conducted at standard conditions and at −65° ±4°F. For tests at low temperatures, the specimen shall be cooled with dry ice. At least twelve specimens shall be tested at standard conditions, half of them being tested in the lengthwise direction and half in the crosswise direction. At least six specimens shall be tested at low temperature in the strongest direction of the sheet. Test results shall be reported as the average in each direction in foot-pounds per inch of specimen width at the gage sections. Location of failure for all specimens shall be reported. If failure of a specimen does not occur in the gage section, an additional test shall be conducted. Any other tensile impact test grip can be used only after approval is granted by the procuring activity.

* 4.6.6 *Tensile tests.* Tensile strength properties shall be determined in accordance with Method 1011 of Standard FED–STD–406. Water immersion shall be for 7 days ±5 hours in distilled water at 73.5° ± 2°F. Specimens shall be tested at standard conditions immediately after removal from water. Specimens for elevated temperature tests shall be conditioned in an air-circulating oven for 8 days ±5 hours and tested at exposure temperature. At least five specimens for determination of tensile strength at standard conditions and after water immersion shall have a configuration similar to the flat tension specimen of Type I as specified in Method 1011 of Standard FED–STD–406. Test specimens for determination of tensile strength at elevated temperatures may be modified to accommodate the test oven and fixtures of clamps for holding the specimen.

* 4.6.7 *Flexural tests.* Flexural test specimens required for Type I material only shall be 1 inch wide and 4 inches long, and the full thickness of the sheet material. All tests shall be conducted over a 2-inch span with center loading. At least six flexural specimens shall be machined, half in the lengthwise and half in the crosswise directions. The maximum load, supported by each specimen taken in the same direction from the same sheet, shall be averaged and compared with the requirements of this specification. Flexural tests at 160° ± 4°F and 275° ± 4°F shall be conducted in an air-circulating oven at exposure temperature. The test apparatus used and rate of loading shall conform to Method 1031 of Standard FED–STD–406.

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* **4.7 Packaging, packing, and marking.** Preparation for delivery shall be examined for conformance to Section 5.

5. PREPARATION FOR DELIVERY

* **5.1 Packaging.** Unless otherwise specified, no packaging is required.

** Gunfire tests have been deleted.

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* 5.2 **Packing.** The levels of packing shall be A, B, or C as specified in the contract or order, in accordance with Specification MIL–P–7936.

* 5.3 **Shipping.** In shipping backing material, including contoured parts, the containers shall be constructed to provide adequate support and preclude physical damage due to impact and shock during handling and transit. All sheets and parts shall be interleaved with paper to minimize scratching or other surface damage, and to prevent sticking to each other.

* 5.4 **Marking.** In addition to any special marking required by the contract or order, marking shall be in accordance with Standard MIL–STD–129.

6. NOTES

* 6.1 **Intended use.** This plastic material is intended for use as backing material for self-sealing fuel and oil tanks such as found in Specification MIL–T–5578, and for non-self-sealing tanks such as found in Specification MIL–T–6396.

* 6.1.1 Class 1 material may be exposed to water, aircraft fuel, or gasoline at temperatures from −65° F to 160° F. Class 2 material may be exposed to water or oil at temperatures from −65°F to 275°F.

6.1.2 Type I or III materials may be used in place of Type II material. Any other interchangeability of types shall not be allowed.

* 6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size and form of plastic backing material required.

(c) Quantity required.

(d) Applicable levels of packaging and packing required (see Section 5).

(e) Installation tests (when desired).

* 6.3 **Changes from previous issue.** The outside margins of this document have been marked "*" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

* 6.4 **Transportation description**

6.4.1 Transportation descriptions and minimum weight applicable to this commodity are:

Rail: Carload minimum weight 30,000 pounds

Motor: Truckload minimum weight 30,000 pounds

* **Military Custodians:**
Army—MR
Navy—WP
Air Force—11

* **Review Activities:**
MR, MO, WP, 11, 69

* **User Activities:**
GL, MC

Preparing Activity:
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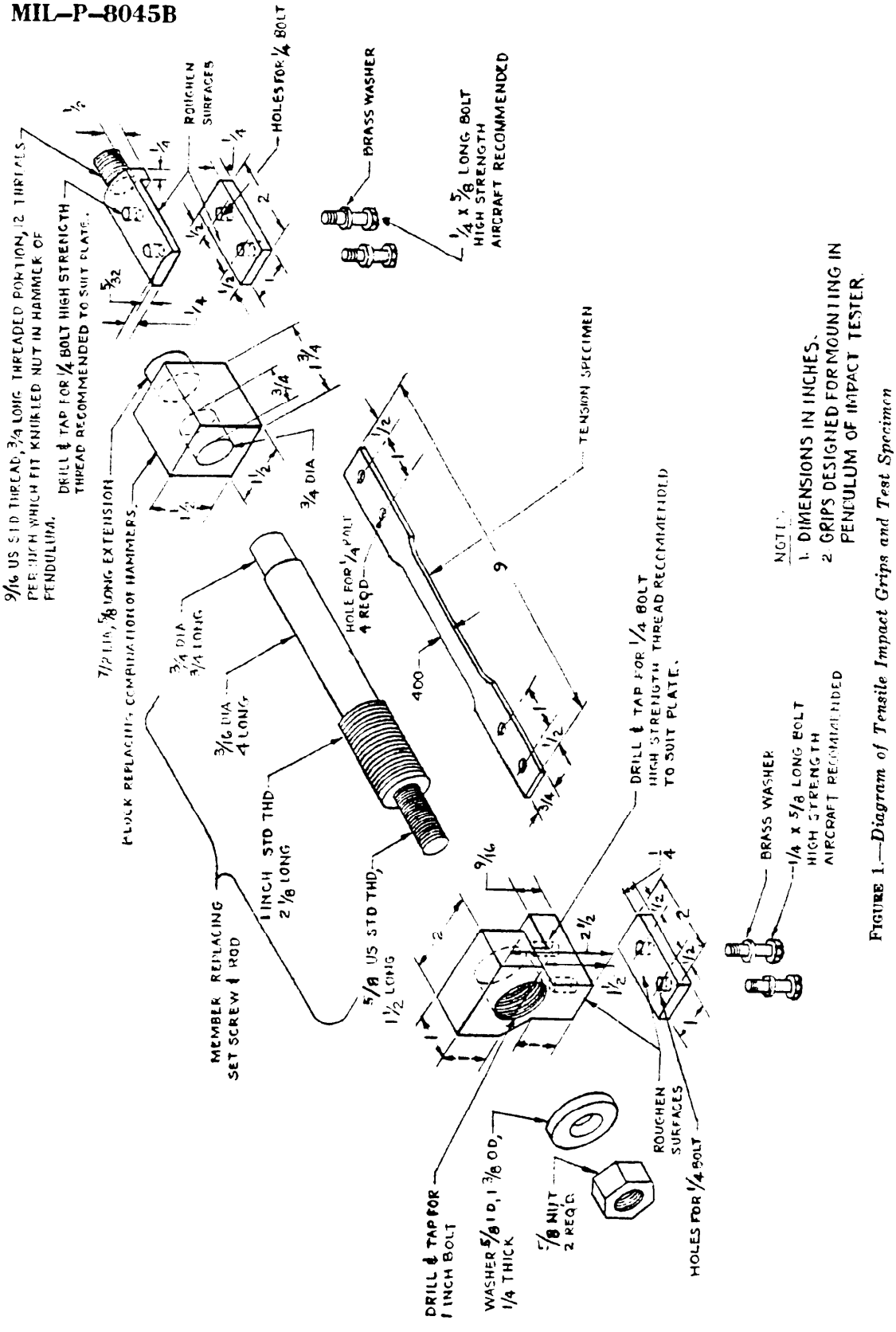


FIGURE 1.—*Diagram of Tensile Impact Grips and Test Specimen*

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